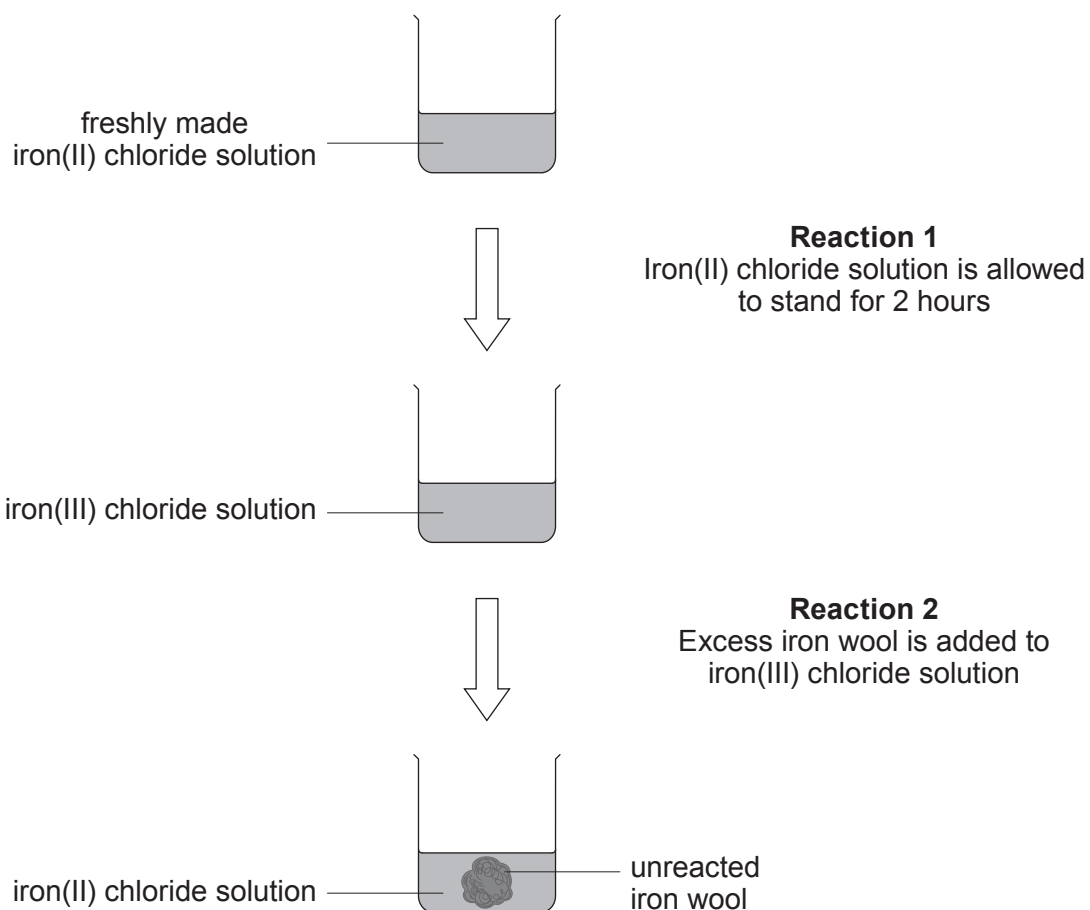


9. Transition metals have the ability to form ions with different charges. Iron and copper are transition metals.

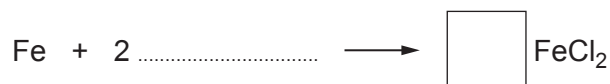
- (a) A teacher carried out a series of reactions to show that iron can form Fe^{2+} and Fe^{3+} ions.



- (i) State what you would expect to **see** during the reactions that shows iron is a transition metal. [1]

- (ii) In **reaction 2**, iron reacts with iron(III) chloride forming iron(II) chloride, FeCl₂.

- I. Complete and balance the equation for this reaction. [2]



- II. Explain the meaning of the term *oxidation* in relation to **reaction 2**. [2]

- (b) Name the reagent used to identify copper(II) ions in solution. Give the observation expected. [2]

Reagent

Observation

